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Co-Designing TWT Scheduling and RF Energy Harvesting for Sustainable Heterogeneous Wi-Fi

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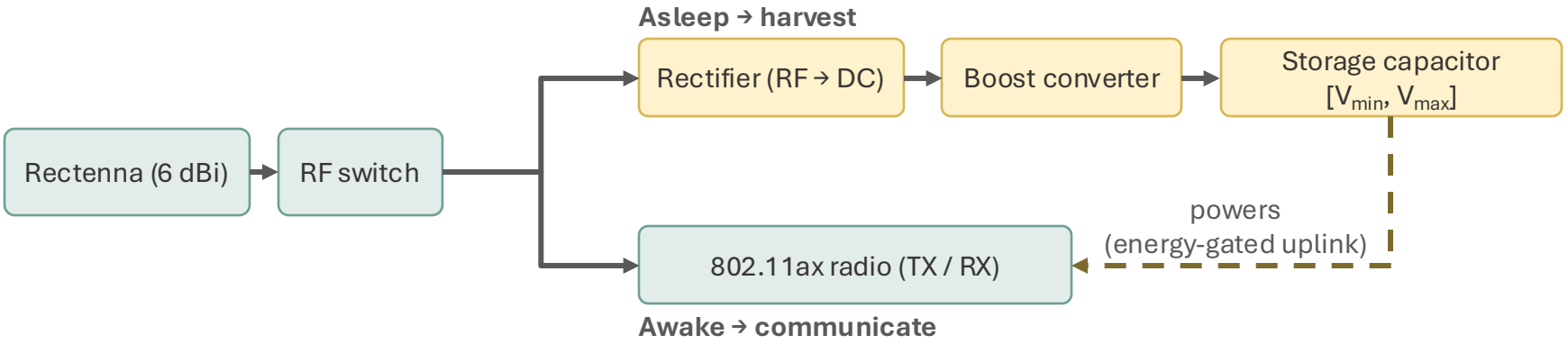
Outline

1. Motivation
2. TWT and the Power Delivery Window
3. Challenge
4. System and harvesting model
5. Problem formulation: state, action, reward
6. Simulation platform
7. RL model and parameters
8. Training and evaluation results
9. Conclusion

Motivation: RF Energy Harvesting Devices (REHD) need intentional MAC design

- Device Type: Wireless sensor networks, Light-weight IoT, Asset tags, etc.
- Plug and Play: minimal infrastructure & maintenance
- Application (Realtime spatio-temporal data): Robotics, Security, Manufacturing, swarm network, etc.

- Harvesting physics completely changes the game
- Harvested energy is very precious and resource hungry
- Cannot spare energy for contention
- Need dedicated power delivery window (PDW)
- Needs MAC level support – TWT+PDW



Time-switching harvester: harvest or communicate, never both.

TWT fits regular devices *and* harvesters

Wi-Fi 6 Target Wake Time (TWT): the AP slices each beacon interval into non-overlapping service periods (SPs). Each STA group wakes only in its own SP.

Harvest or TX/RX, never both

A time-switching rectenna either harvests or powers the radio. TWT gives it a clean sleep/wake schedule.

Contention-free access

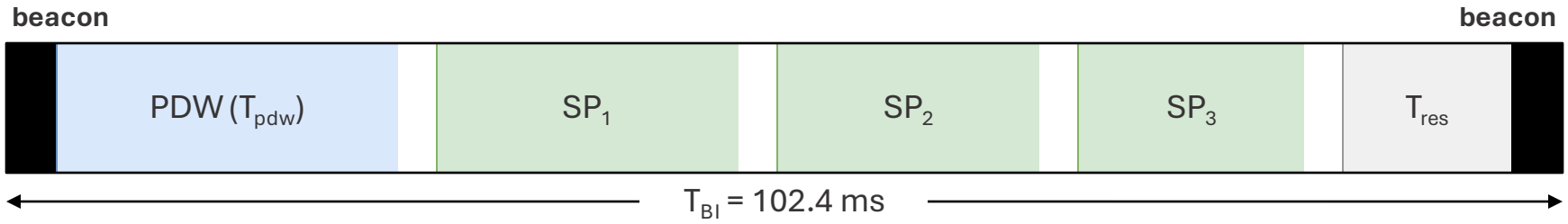
A dedicated SP spares REHDs the energy cost of contending with high-throughput STAs.

The gap

TWT research optimizes QoS only (grouping, fairness, dynamic traffic) and ignores energy harvesting. DRL for Wi-Fi (OFDMA resource units, application-layer QoS) uses simplified channels and never couples traffic with energy delivery.

IEEE 802.11ax-2021; Venkateswaran et al., Comput. Commun. 2025 and WNS3 2024; Chen & Zhu, IEEE TWC 2020; Tan et al., ICTC 2024; Li et al., VTC 2025-Spring

Co-design: TWT schedule + PDW



Power Delivery Window (PDW)

- Right after the beacon, the AP sends dummy packets for duration of T_{pdw}
- Sleeping REHDs harvest them; other STAs, asleep outside their SP, ignore them

The tension

- Longer PDW → more energy, but less airtime for uplink/downlink, and vice versa

$$\Sigma_k d_k + (K - 1)\cdot\delta_g \leq T_{BI} - T_{res} - T_{pdw}$$

Challenge: joint TWT + PDW scheduling is NP-hard

Every 25 beacon intervals, the AP must jointly choose:

1. TWT schedule (K groups, wake durations and offsets)
2. STA-to-group assignment
3. PDW length

- 20 STAs and up to 8 groups: the assignment alone has $8^{20} \approx 1.2 \times 10^{18}$ options; durations and PDW multiply it
- Even the TWT grouping sub-problem alone is NP-hard; no closed-form optimum
- Traffic is non-stationary, and the AP is class-blind
- Energy delivered now pays off only several steps later, so heuristics cannot help

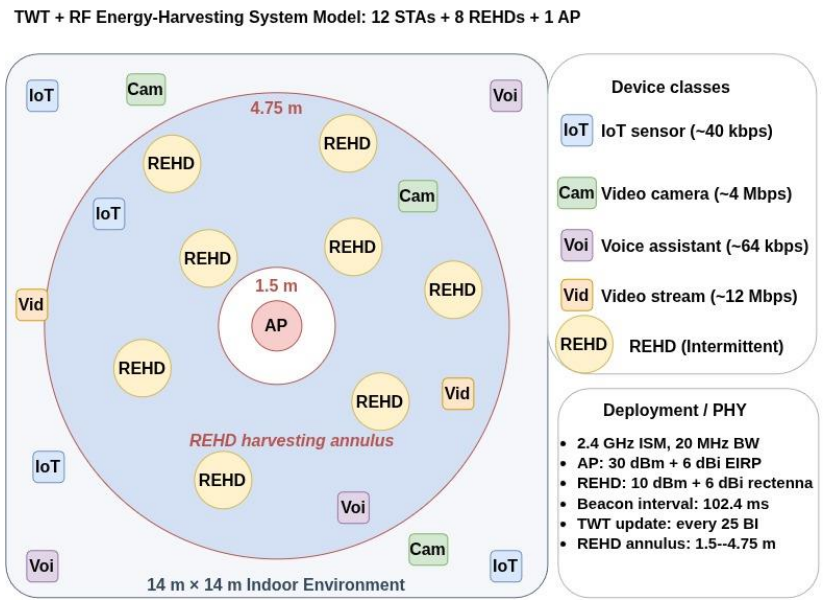
Propose: one DRL agent that co-designs TWT and energy delivery

No existing work offers all of: datasheet PowerCast harvesters in full-stack ns-3 · an AP-controlled Power Delivery Window · SPs + grouping + PDW in one decision · protocol-compliant, class-blind observations.

Question: Can the AP learn, from protocol-compliant signals only, a joint TWT schedule, grouping and PDW that keeps battery-free harvesters alive while serving heterogeneous traffic fairly?

Puligheddu et al., IEEE PIMRC 2024; Maksud & Carvalho, ICCCN 2026

System model: 1 AP + 20 heterogeneous STAs

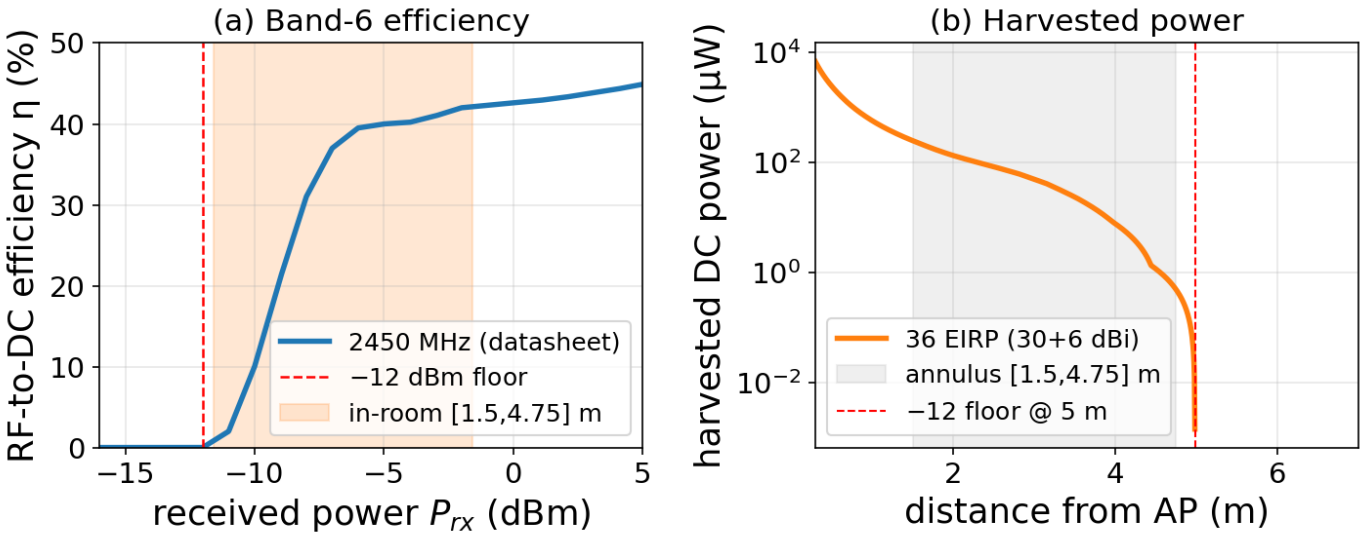


Example with 8 REHDs in the harvesting annulus.

- Single 802.11ax AP
- Unilateral TWT announced in beacons (no per-STA negotiation)
- 20 STAs in a 14 m × 14 m room on 2.4 GHz; 4–16 of them are REHDs
- Conventional classes: IoT 40 kbps · voice 64 kbps · camera 4 Mbps · video 12 Mbps
- REHDs: four PowerCast hardware sub-types, 1.5–4.75 m from the AP
- Packet deadlines: REHD 200 ms, others 0.5–3 s; expired packets are dropped

The AP is class-blind: it must infer who is energy-bound from how STAs respond to its actions.

PowerCast model: harvesting reality



Link budget

$P_{rx} = P_{EIRP} - PL(d) + G_{rect}$

Friis path loss;

36 dBm EIRP (30 dBm + 6 dBi);

6 dBi rectenna.

Datasheet efficiency

$E_{harv} = \eta_b \cdot \eta(P_{rx}) \cdot P_{rx} \cdot \Delta t$

η is 0 below −12 dBm and peaks near 46 %;

boost converter $\eta_b = 0.85$.

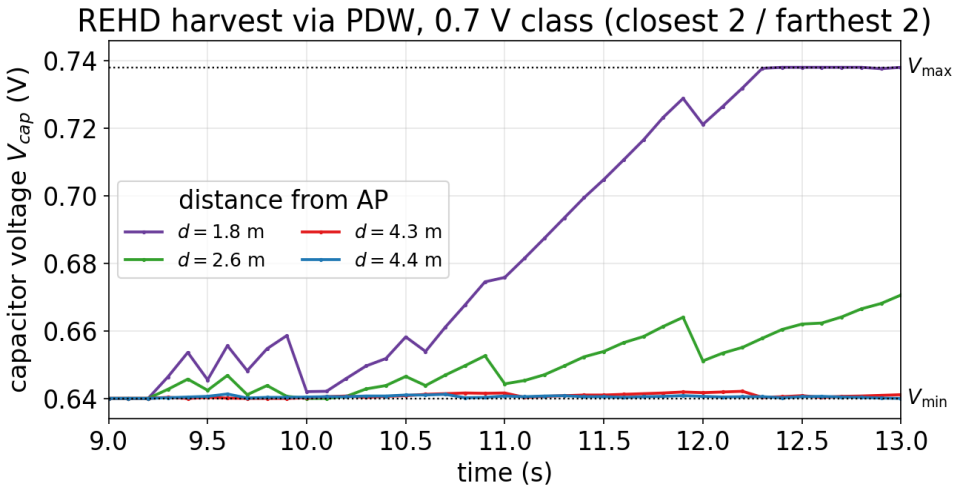
In the room

P_{rx} is about −1.6 to −11.6 dBm across the annulus, giving ~250 µW down to ~0.5 µW.

No harvest beyond ~5 m.

PowerCast P21XXCSR-EVB datasheet

Stored charge gates every uplink packet



Capacitor voltage of four REHDs (0.64–0.738 V class) at different distances.

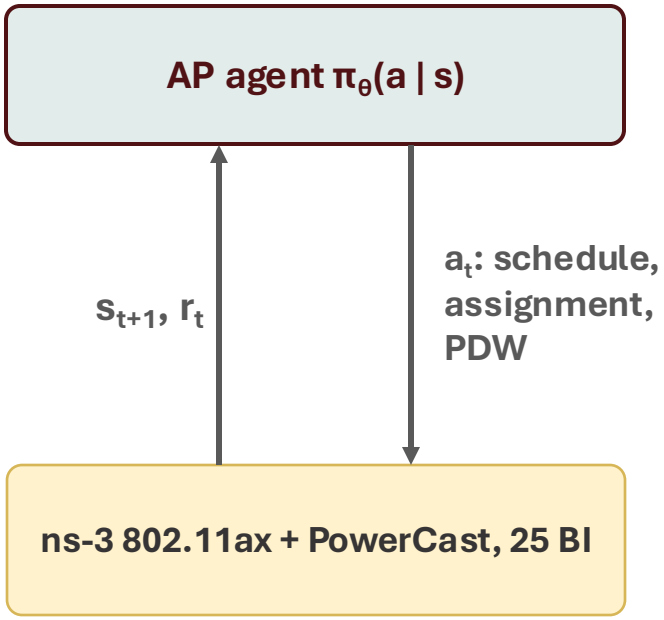
- Capacitor voltage is confined to:

  $[V_{min}, V_{max}]$; harvesting stops at V_{max}
- REHDs harvest only while asleep (time-switching)
- Energy-gated uplink: a REHD sends only if stored energy above V_{min} covers the packet
- Radio consumption:

  ≈ 13.3 mW TX, 7.5 mW RX, 1 µW sleep
- Near REHD (1.8 m) climbs to V_{max}
- Far REHDs (~4.4 m) stay pinned at V_{min}

Problem formulation: an episodic MDP

Element	Definition
State s_t	$s_t \in \mathbb{R}^{7N}$: 7 realistic per-STA features (N = 20)
Action a_t	$(a_{sched}, a_{assign}, a_{pdw}) \in \mathcal{A}_s \times \mathcal{A}_m \times \mathcal{A}_p$; 24 × 25 × 10
Transition P	ns-3 runs Δ_{step} = 25 BI (2.56 s) of full 802.11ax MAC/PHY plus energy dynamics
Reward r_t	α-fair QoS utility minus a harvester-expiry penalty (slide 13)
γ, horizon	γ = 0.99; 100 decisions per episode
Objective	$\max_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi\theta}[\Sigma_t \gamma^t r_t]$



Every action is decoded to satisfy $\Sigma_k d_k + (K-1)\cdot\delta_g \le T_{BI} - T_{res} - T_{pdw}$;
the SPs, with 2 ms guards, fit between the end of the PDW and 95 ms of the 102.4 ms beacon interval.

Observation space: realistic vs oracle

$s_t \in \mathbb{R}^{7N}$: 7 realistic features per STA (N = 20)

Realistic → actor (deployed)

- Buffer backlog (BSR)
- Packets received
- Airtime used
- RX failures (FCS)
- SNR
- Silence (time since last RX)
- SP starvation

Per-step levels, deltas and ratios;
no device class;
log-scaled, clipped to [0, 1], z-scored

Oracle QoS → reward

transmitted · enqueued · expired packets

Oracle energy → critic (training only)

state of charge · harvested · consumed · is-REHD

Simulator only: never seen at deployment

Action space: $>10^{18}$ → 6,000

24

Schedule head

$K \in \{2, 3, 4, 5, 6, 8\}$ groups × 4 wake budgets

Equal SPs, 2 ms guards; 36–84 ms of SPs per BI

×

25

Assignment head

Round-robin (4)

Contiguous split into 2–10 blocks (9)

Tail cut (12): a block of 4–14 STAs gets its own group

×

10

PDW head

T_{pdw} = 0, 5, …, 45 ms

SPs are compressed into the rest of the BI

= 6,000 joint actions

Three independent categorical heads, carved from an EDA sweep (dead and dominated actions pruned)

Reward function: α-fair QoS utility

$$r_t = U_\alpha(q) - w_e \cdot \bar{e}_{REHD}, \quad w_e = 0.5$$

Per-STA satisfaction

$q_i = \text{served}_i \cdot (1 - \text{loss}_i)$

served = transmitted ÷ admitted,
loss = expired ÷ admitted,
+3 smoothing (an idle STA counts as satisfied)

Fair aggregate (α = 1)

$U_\alpha(q) = (1/N)\ \Sigma_i \log q_i$

Concave: a starved harvester can't be averaged away

Harvester penalty

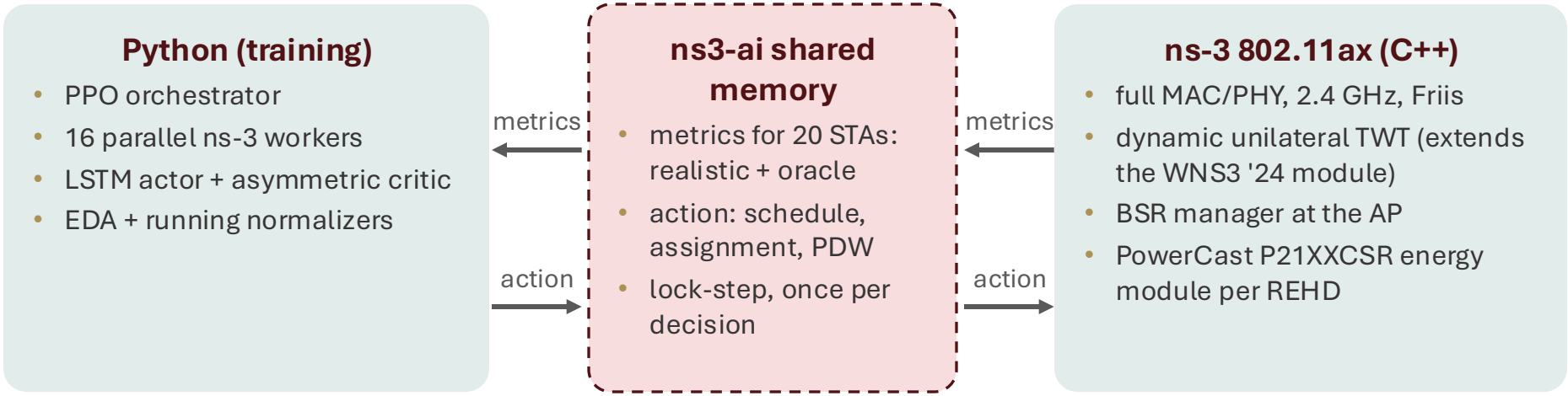
$\bar{e}_{REHD}$ = mean over REHDs of Δexp ÷ (Δexp + Δtx)

Punishes harvester packets that expire

No energy term: an energy-locked harvester can't transmit, so the QoS terms already punish starving it. Energy state goes only to the critic.

Kelly et al., J. Oper. Res. Soc. 1998; Mo & Walrand, IEEE/ACM ToN 2000

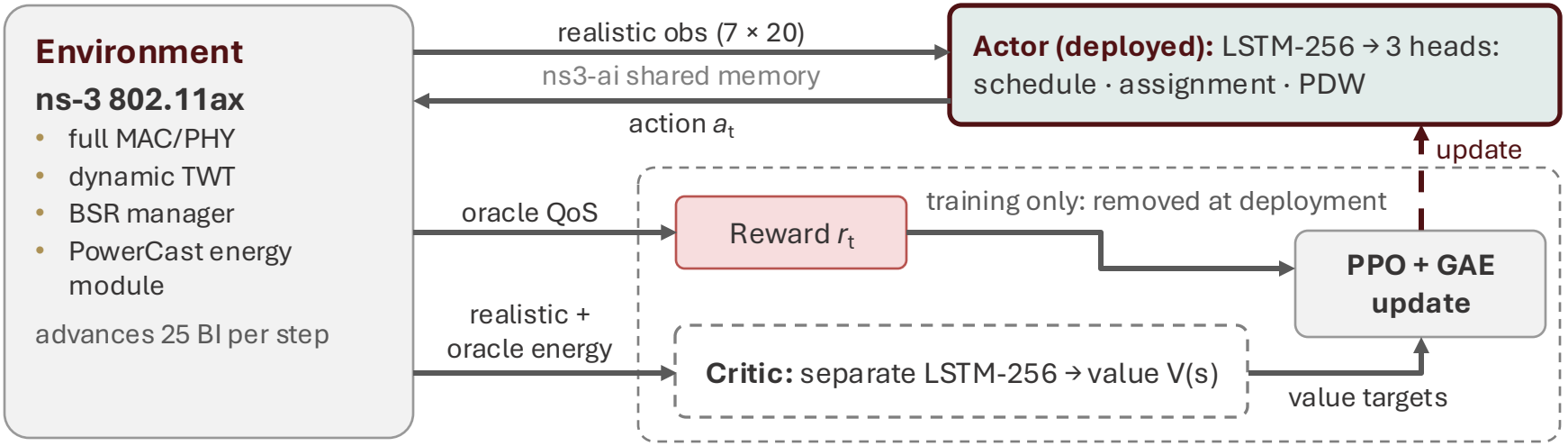
Simulation platform: ns-3 + ns3-ai + Python



- 1 step = 25 BI (2.56 s); 1 episode = 9.2 s start-up + 100 steps ≈ 265 s simulated, each in its own spawned process
- 5 traffic segments per episode, randomly over- or under-saturated; 4–16 REHDs of 20 per scenario
- Each batch: 24 episodes in 4 common-random-number groups (same scenario, different sampled actions)

Yin et al. (ns3-ai), WNS3 2020; Venkateswaran et al., WNS3 2024

Recurrent actor + asymmetric critic



Why recurrent?

A drained harvester and a congested STA look alike in realistic signals but need opposite actions: more sleep to charge vs. more airtime. The LSTM integrates evidence over time.

Why asymmetric?

Energy delivered now becomes served uplink several steps later. A critic that also sees capacitor state can credit "harvest now → serve later". It is used ***only*** for training.

Pinto et al., 2017; Lowe et al., NeurIPS 2017; Schulman et al., PPO 2017 and GAE 2015; Yin et al. (ns3-ai), WNS3 2020

Model and training parameters

Networks

	Actor (deployed)	Critic (training only)
Input	140 realistic (7 × 20)	220 (+ 4 × 20 oracle energy)
Recurrent core	LSTM, 1 layer, 256	separate LSTM, 1 layer, 256
Trunk	2 × (Linear 256 + Tanh)	2 × (Linear 256 + Tanh)
Output	3 categorical heads: 24 × 25 × 10	value V(s)
Parameters	≈ 0.55 M	≈ 0.62 M

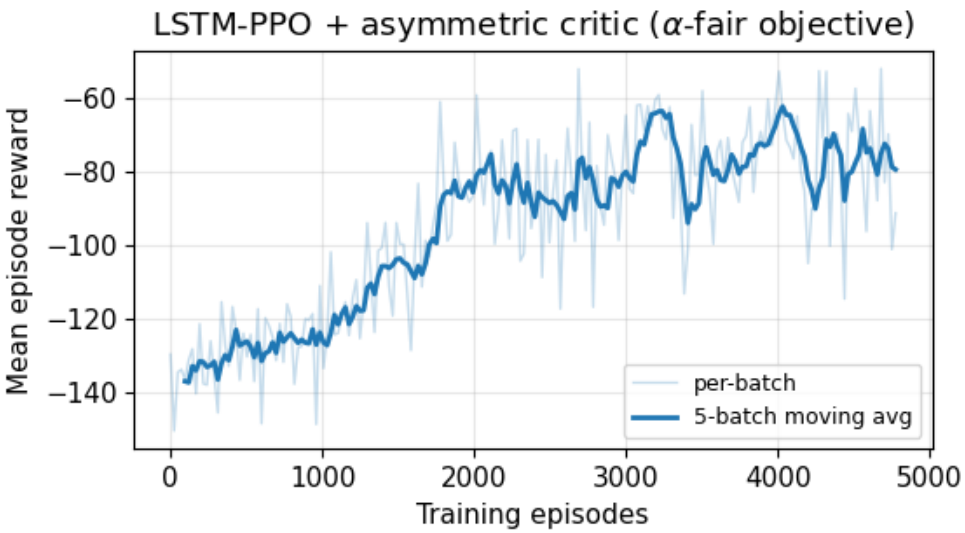
Orthogonal initialization of the linear layers (gain $\sqrt{2}$);
running z-score on inputs, warm-started from the EDA;
a separate normalizer for the oracle block.

PPO

Setting	Value
Learning rate	3×10^{-4}
Epochs per update	10
γ / λ_{GAE}	0.99 / 0.95
Clip range	0.2
Entropy / value coef.	0.01 / 0.5
Max grad-norm	0.5
Normalization	advantages + returns

Training budget: 200 updates × 24 episodes = 4,800 episodes (460,800 decisions) on 16 parallel ns-3 workers, ≈ 14.5 h wall-clock

Training and evaluation setup



Mean episode reward during training
(LSTM-PPO + asymmetric critic, α-fair objective).

Training

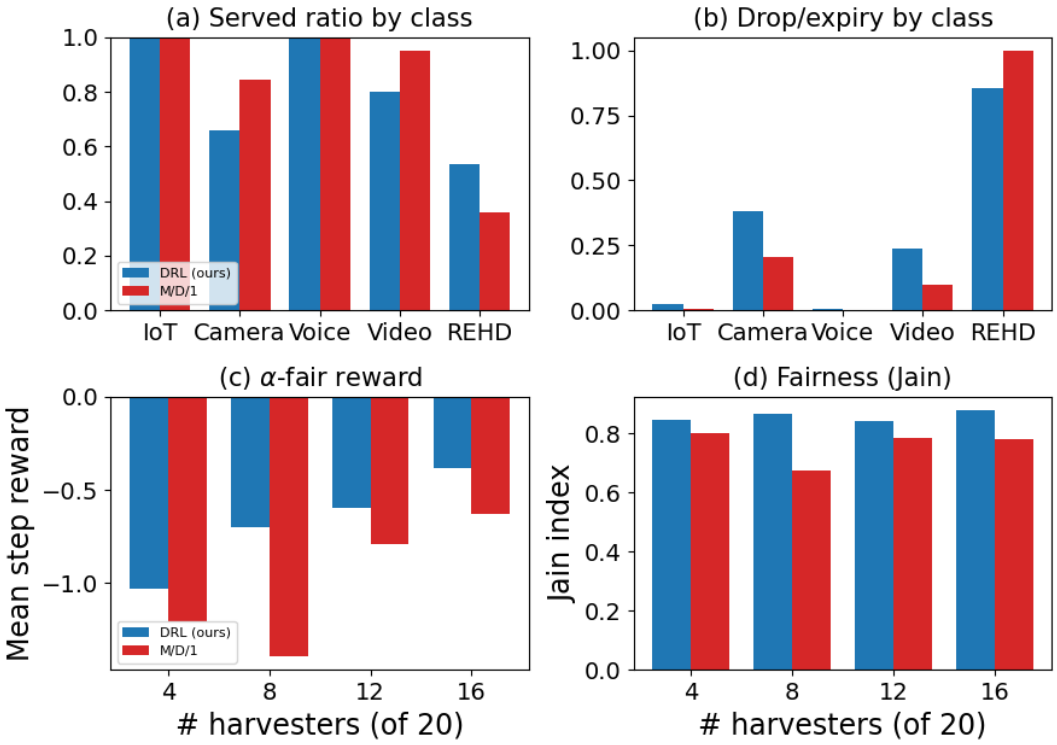
- 4,800 training episodes; the reward climbs, then plateaus
- At deployment, the actor runs alone: no critic, no oracle signals

Evaluation

- Baseline: online M/D/1 queueing scheduler from our ICCCN '26 work, plus an energy-aware PDW rule. It uses the same realistic inputs and the same action space.
- 4, 8, 12 or 16 REHDs of 20 STAs × 20 matched seeds × 2 schedulers = 160 episodes

Schulman et al., 2017; Andrychowicz et al., 2020; Shortle et al., Fundamentals of Queueing Theory, 2018

Result: Harvesters protected



Blue = DRL (ours), red = M/D/1. (a, b) per class; (c, d) vs. number of harvesters.

A queueing optimizer drains the high-rate STAs; our α-fair agent shifts that airtime to energy-starved harvesters.

- (a, b) REHD served ratio 0.36 → 0.54; REHD drop/expiry 1.00 → 0.86
- (a, b) The cost lands on the high-rate camera and video classes
- (c) Higher α-fair reward with 4, 8, 12 and 16 harvesters
- (d) Jain fairness 0.84–0.88 vs. 0.68–0.80

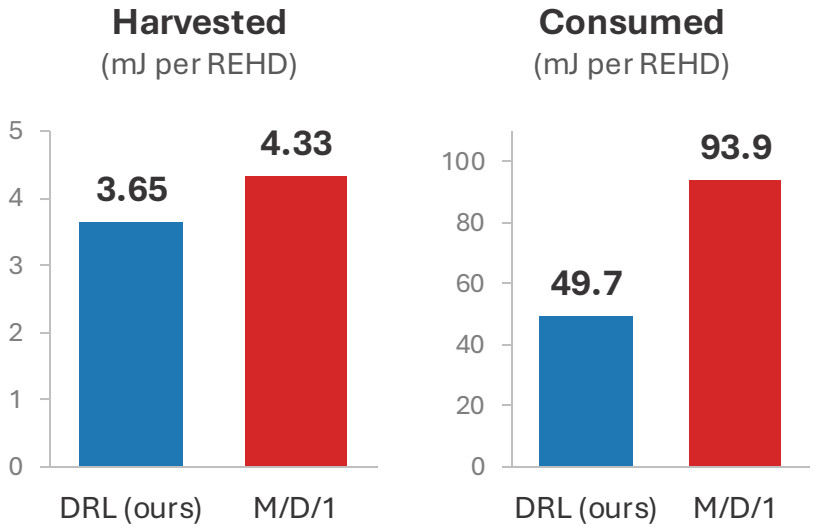
+0.33 better α-fair welfare per step (−0.68 vs. −1.00); better in 89 % of scenarios

+0.11 better Jain fairness (0.88 vs. 0.77); better in 88 % of scenarios

Cost to protect harvesters:
Served ratio 0.875 vs. 0.926
Throughput 147.6 vs. 195.3 KB/STA-step
Drop/expiry 0.167 vs. 0.117
Latency 771 vs. 747 ms

Served = packets transmitted ÷ packets admitted to the STA's MAC queue (the demand an AP can observe).

Result: 2.8× more harvester demand served per joule



1.9× less

energy consumed by harvesters (49.7 vs. 93.9 mJ)

2.8× more

harvester demand served per joule (10.8 vs. 3.8)

Harvester-light (4 REHDs): M/D/1 spends 385 mJ to serve 4 % of harvester demand; our agent serves 40 % with 117 mJ.

Slightly less energy harvested, but spent on useful uplink instead of keeping harvesters awake in contention.

Served = packets transmitted ÷ packets admitted to the STA's MAC queue (the demand an AP can observe).

Conclusion

1. **Co-design works.** One AP policy sets the TWT schedule, STA grouping and a Power Delivery Window.

2. **Realistic and deployable.** A datasheet PowerCast model in full-stack ns-3, with a class-blind, protocol-compliant actor.

3. **Asymmetric critic.** It solves the delayed "harvest now → serve later" credit; the deployed actor stays realistic.

4. **vs. M/D/1.** 1.9× less harvester energy, 2.8× more harvester demand per joule, +0.11 Jain, and an α-fair win in 89 % of scenarios, for ~5 % less served demand.

Next steps: Hardware validation with PowerCast P21XXCSR boards · multi-AP / overlapping BSSs · joint TWT–OFDMA–MLO scheduling with energy delivery

Thank you!
Questions?

github.com/TXST-SHINe-Lab/
PowerCast-TWT-DRL-DynamicTraffic--MobiArch26

